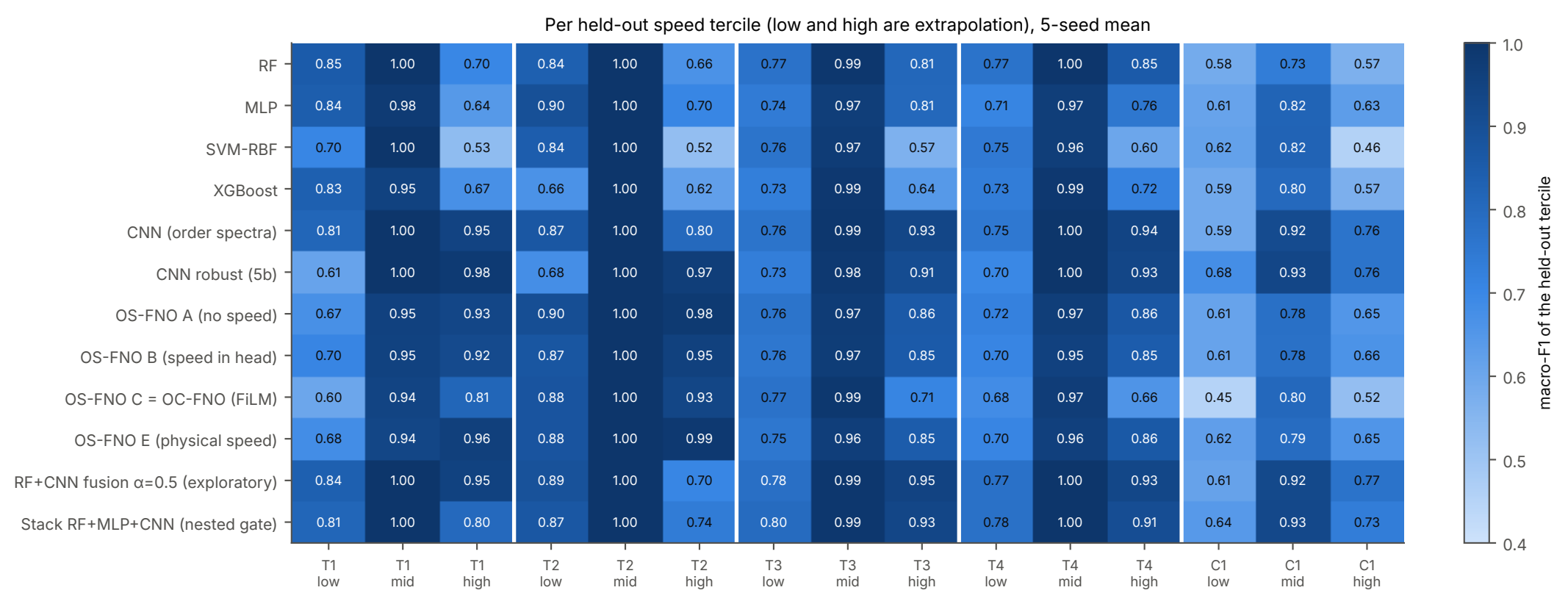
Per held-out speed tercile (low and high are extrapolation), 5-seed mean
RF 0.85 1.00 0.70 0.84 1.00 0.66 0.77 0.99 0.81 0.77 1.00 0.85 0.58 0.73 0.57
MLP 0.84 0.98 0.64 0.90 1.00 0.70 0.74 0.97 0.81 0.71 0.97 0.76 0.61 0.82 0.63
SVM-RBF 0.70 1.00 0.53 0.84 1.00 0.52 0.76 0.97 0.57 0.75 0.96 0.60 0.62 0.82 0.46
XGBoost 0.83 0.95 0.67 0.66 1.00 0.62 0.73 0.99 0.64 0.73 0.99 0.72 0.59 0.80 0.57
CNN (order spectra) 0.81 1.00 0.95 0.87 1.00 0.80 0.76 0.99 0.93 0.75 1.00 0.94 0.59 0.92 0.76
CNN robust (5b) 0.61 1.00 0.98 0.68 1.00 0.97 0.73 0.98 0.91 0.70 1.00 0.93 0.68 0.93 0.76
OS-FNO A (no speed) 0.67 0.95 0.93 0.90 1.00 0.98 0.76 0.97 0.86 0.72 0.97 0.86 0.61 0.78 0.65
OS-FNO B (speed in head) 0.70 0.95 0.92 0.87 1.00 0.95 0.76 0.97 0.85 0.70 0.95 0.85 0.61 0.78 0.66
OS-FNO C = OC-FNO (FiLM) 0.60 0.94 0.81 0.88 1.00 0.93 0.77 0.99 0.71 0.68 0.97 0.66 0.45 0.80 0.52
OS-FNO E (physical speed) 0.68 0.94 0.96 0.88 1.00 0.99 0.75 0.96 0.85 0.70 0.96 0.86 0.62 0.79 0.65
RF+CNN fusion α=0.5 (exploratory) 0.84 1.00 0.95 0.89 1.00 0.70 0.78 0.99 0.95 0.77 1.00 0.93 0.61 0.92 0.77
Stack RF+MLP+CNN (nested gate) 0.81 1.00 0.80 0.87 1.00 0.74 0.80 0.99 0.93 0.78 1.00 0.91 0.64 0.93 0.73
T1 low
T1 mid
T1 high
T2 low
T2 mid
T2 high
T3 low
T3 mid
T3 high
T4 low
T4 mid
T4 high
C1 low
C1 mid
C1 high
macro-F1 of the held-out tercile
1.0
0.9
0.8
0.7
0.6
0.5
0.4